

Analytical Apps

With the ABAP Core Data Services (CDS), you can develop and extend analytical, lifecycle-stable and upgrade-safe apps and services.

▼ About Analytical Apps

With analytical apps, you can build sophisticated and complex analytical data models to evaluate and analyze business data in your ABAP system in two scenarios:

- **Embedded Analytics:** In embedded analytics, the ABAP analytical engine is part of the software stack and operates on the same data as the transactional apps that maintain the business data. The analytical evaluations operate directly on the business data, without data replication to an external data warehouse system. Instead, the real-time business data is queried to always evaluate the most recent changes and trends in your business data.
- **Side-by-Side Analytics:** In a side-by-side analytics scenario, the business data is replicated from an analytical source system to other analytical clients (like SAP Datasphere) or to a data warehouse. The business data is replicated from an ABAP system based on a predefined integration scenario to allow seamless data exchange.

ABAP Cloud mainly focuses on embedded analytics, but also enables side-by-side-analytics for example with SAP Analytics Cloud.

Qualities

- Define complex multidimensional models natively with ABAP Core Data Services
- Reuse of development objects across transactional and analytical use cases
- Create services that are optimized for SAP HANA to make full use of all advantages like buffering and performance optimizations
- Create lifecycle-stable multidimensional models that can be extended without modifications
- Develop metadata-driven SAP Fiori UIs with analytics annotations for a consistent user experience

▼ Common Development Patterns

- **Hierarchies:** Enables you to display a parent-child hierarchy to navigate the data set, like for example from displaying the products sales per `region` to display the product sales per `country`.
- **Multidimensional data models:** Business data always has multiple angles you want to analyze. Multidimensional data models allow you to include all important data sources in a typical star schema.
- **Charts and Dashboards:** You can expose your aggregated data. Represent semantic views on business data through the use of specific patterns that are independent of UI technologies. to build charts and dashboards with UI annotations.

▼ Common Annotations

- **Analytics Annotations:** Enable the analytical engine for multidimensional data consumption, performing data aggregation, and slicing and dicing data.

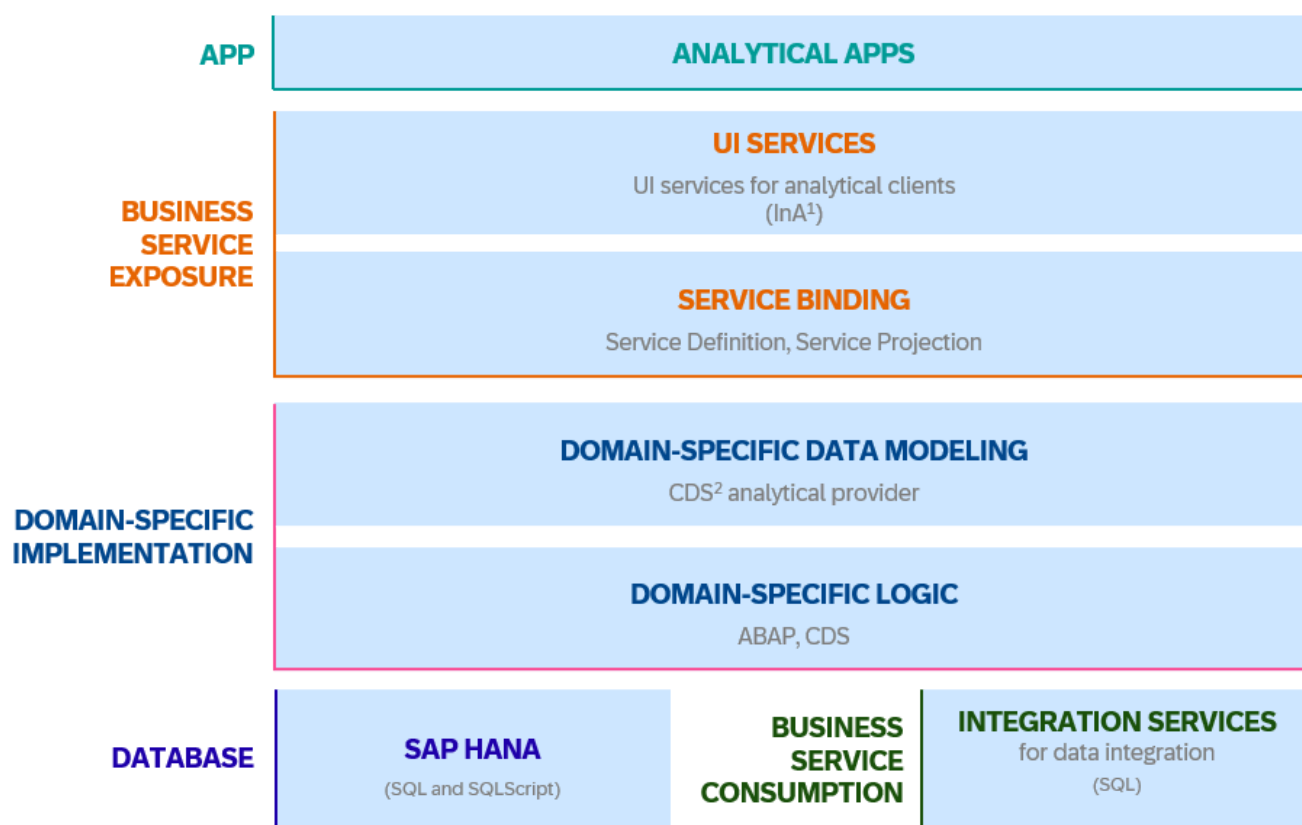
- **AnalyticsDetails Annotations:** Enable you to specify the default multidimensional layout of the query, the sequence of variables in UI consumption, and the specific aggregation and planning behavior of the data.
- **Aggregation Annotations:** Specify aggregation behavior on element level like for example `sum`.
- **Hierarchy Annotations:** Enable you to specify a parent-child hierarchy that they want to make explicitly accessible in a data model, together with the structure that defines this hierarchy.

Architecture

The basis for an analytical app is a multidimensional data model that combines your business KPIs with contextual information. The center of the multidimensional data model is a cube that is joined with several dimensions that contain the contextual business information. The combination of cube, dimensions, Text views , and hierarchy views is called `analytical provider`.

Everything for the multidimensional data model is implemented with ABAP Core Data Services (CDS) in combination with different annotations that optimize the data model for the analytical use case. With the business service projection or analytical query you define the initial layout that is displayed in the analytical client.

Within analytical queries, you define the analytical evaluations, filters, and calculations you want to perform like for example displaying sales per quarter or region. Analytical data models are exposed with the `Information Access` protocol that can then be consumed by different analytical clients.



¹ Information access

² Core Data Services

Develop Analytical Apps

▼ Recommended Readings

- [ABAP Analytics Scenario Overview](#)

- [Analytical Data Modeling](#)
- [Analytical Queries](#)
- [Analytical Use Cases - Initial Considerations \(ABAP Cloud Guide\)](#)
- [Conceptual Background: Access Management \(ABAP Cloud Guide\)](#)

Get Started

If you want to get started with building analytical apps, you can use the [Build Analytical Applications with ABAP Cloud](#) repository. This gives you guidance to develop a simple analytical app from multidimensional data model to analytical query and UI service.

You can also watch [Introduction to Embedded Analytics with ABAP Cloud](#) video to get some insights into how embedded analytics works.

In addition, you can check out the blog posts [Embedded Analytics with ABAP Cloud - A Brief Overview \(Part 1\)](#) and [Embedded Analytics with ABAP Cloud – The Multidimensional Cube \(Part 2\)](#) to learn more about multidimensional data models.

You can also read the chapters [Analytical Data Modeling](#), [Star Schema and Snowflake Schema](#), and [Analytical Queries](#) of the analytics documentation to get the conceptual background for multidimensional data model in ABAP CDS.

Develop

To develop an analytical app, you need a cube, dimensions (optionally with hierarchies), an analytical query and a UI service.

You can find a sample implementation in the [Develop](#) chapter of the documentation. The chapters [Dimensions](#), [Hierarchies](#), and [Cubes](#) introduce the corresponding development objects and which annotations you need to define them.

The chapter [Defining CDS Hierarchies](#) gives you an example of how to implement hierarchies for different use cases.

Extend

Like all CDS-based data models, multidimensional data models can be extended. The chapter [Extend Analytical UI Services \(ABAP Cloud Guide\)](#) gives an overview of all extension possibilities for analytical data models.

Test

You can use the available test double frameworks to develop tests for the multidimensional data model. [Developing Test for Analytical Development Scenarios](#) for more information.

Deploy

Development objects created for an analytical app rely on the respective lifecycle management of the respective products. For more information, see [Lifecycle Management \(ABAP Cloud guide\)](#).

More Information and Support

- [ABAP analytics documentation](#) on the help portal: Contains all the information about multidimensional data models, ABAP analytics and how to model them using ABAP Core Data Services.

- [ABAP resource center for analytical apps](#): Find an overview of all available resources for developing analytical apps.
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